

DRAVA: AN EVENT-DRIVEN RUNTIME FOR ENERGY-EFFICIENT, AGENT-AIDED SCIENTIFIC WORKFLOWS

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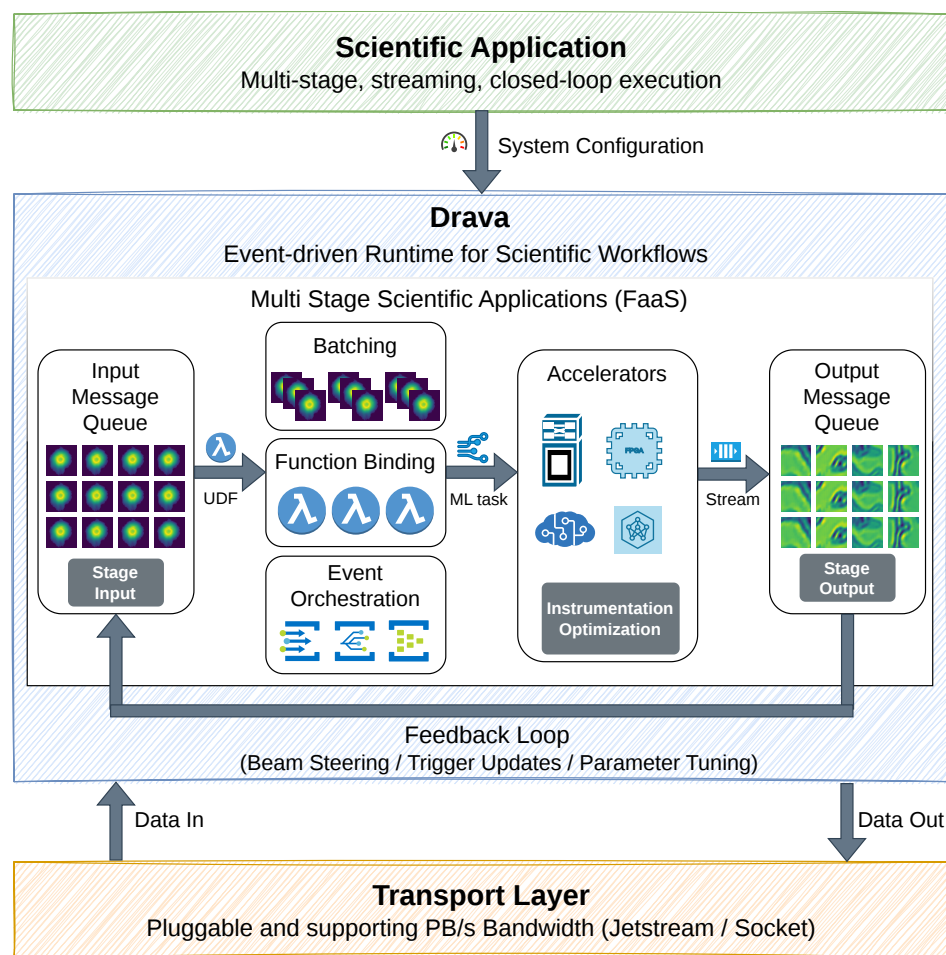
ABSTRACT

Drava is an event-driven runtime for scientific streaming workflows. It uses a lightweight Function-as-a-Service model where application stages are expressed as reusable functions triggered by data events.

Core contributions of Drava are:

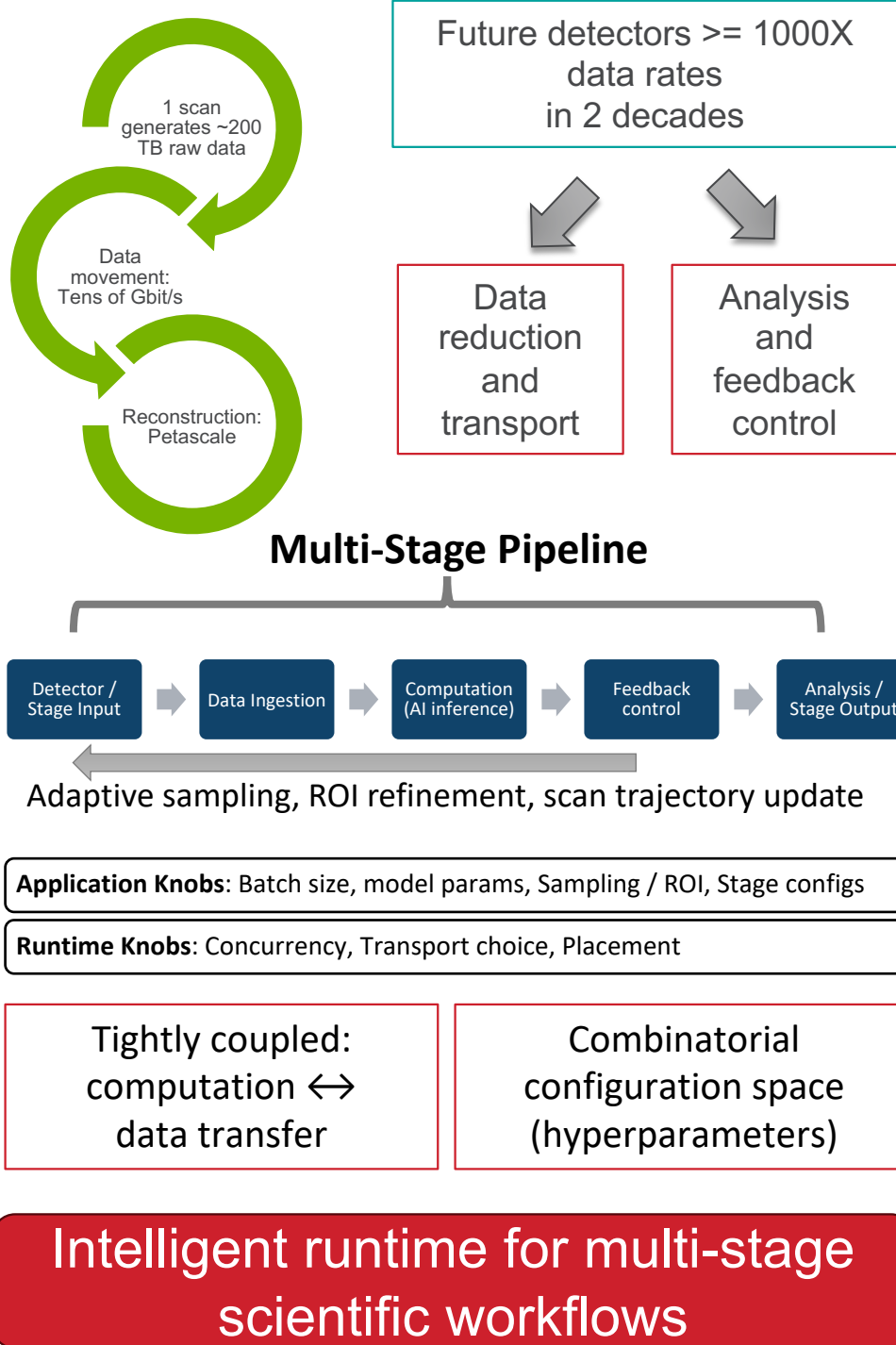
- Event-driven **abstraction**
- Transport-aware **orchestration**
- End-to-end **instrumentation**
- Agentic **optimization**

We evaluate Drava on ptychography and tomography workflows on JLSE, demonstrating high-throughput execution under increasing detector data-rate and bandwidth demands



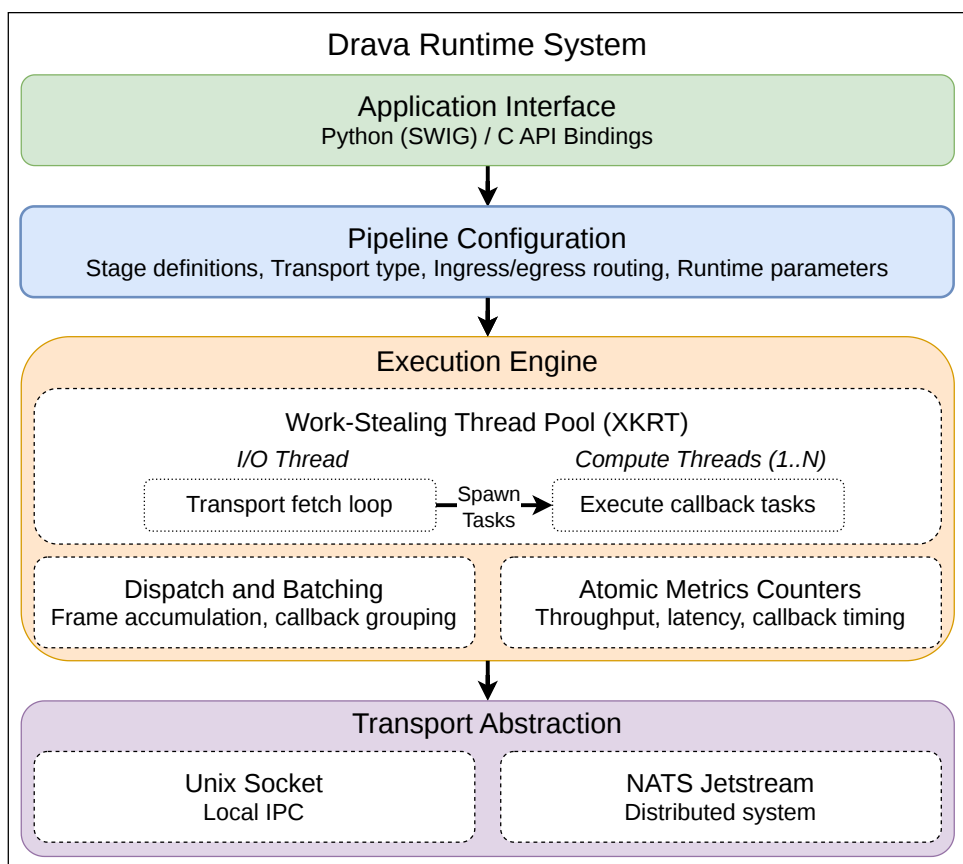
Drava enables event-driven, observable, and optimizable pipelines

MOTIVATION



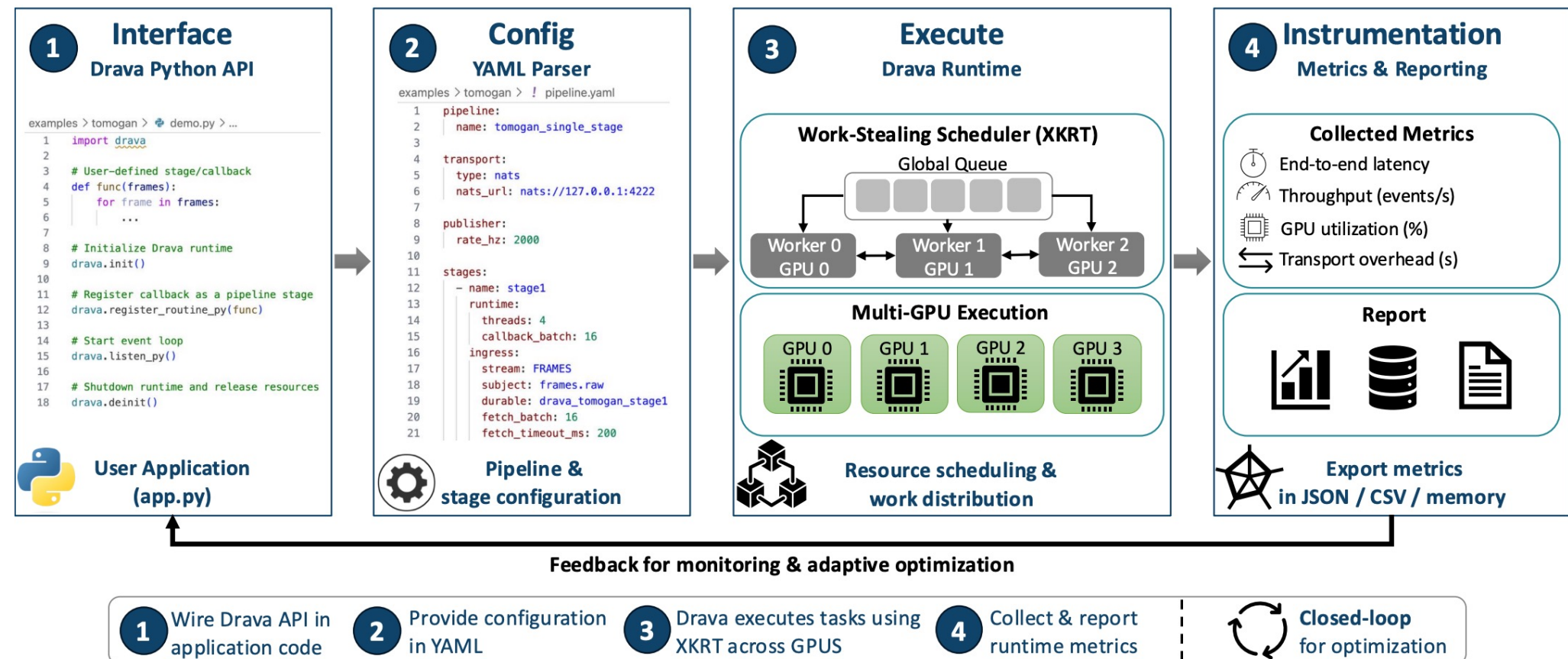
ARCHITECTURE

- Layered abstraction model
- Built-in dispatch and batching
- Pluggable transport layer
- End-to-end performance metrics



APPLICATION PIPELINE AND INSTRUMENTATION

- Drava separates application logic from runtime orchestration. Developers implement stage functions, while the runtime manages communication, batching, scheduling, and metric collection.
- This design allows the same workflow to run under different execution configurations without changing application code. It also enables systematic measurement of how runtime parameters affect end-to-end performance.



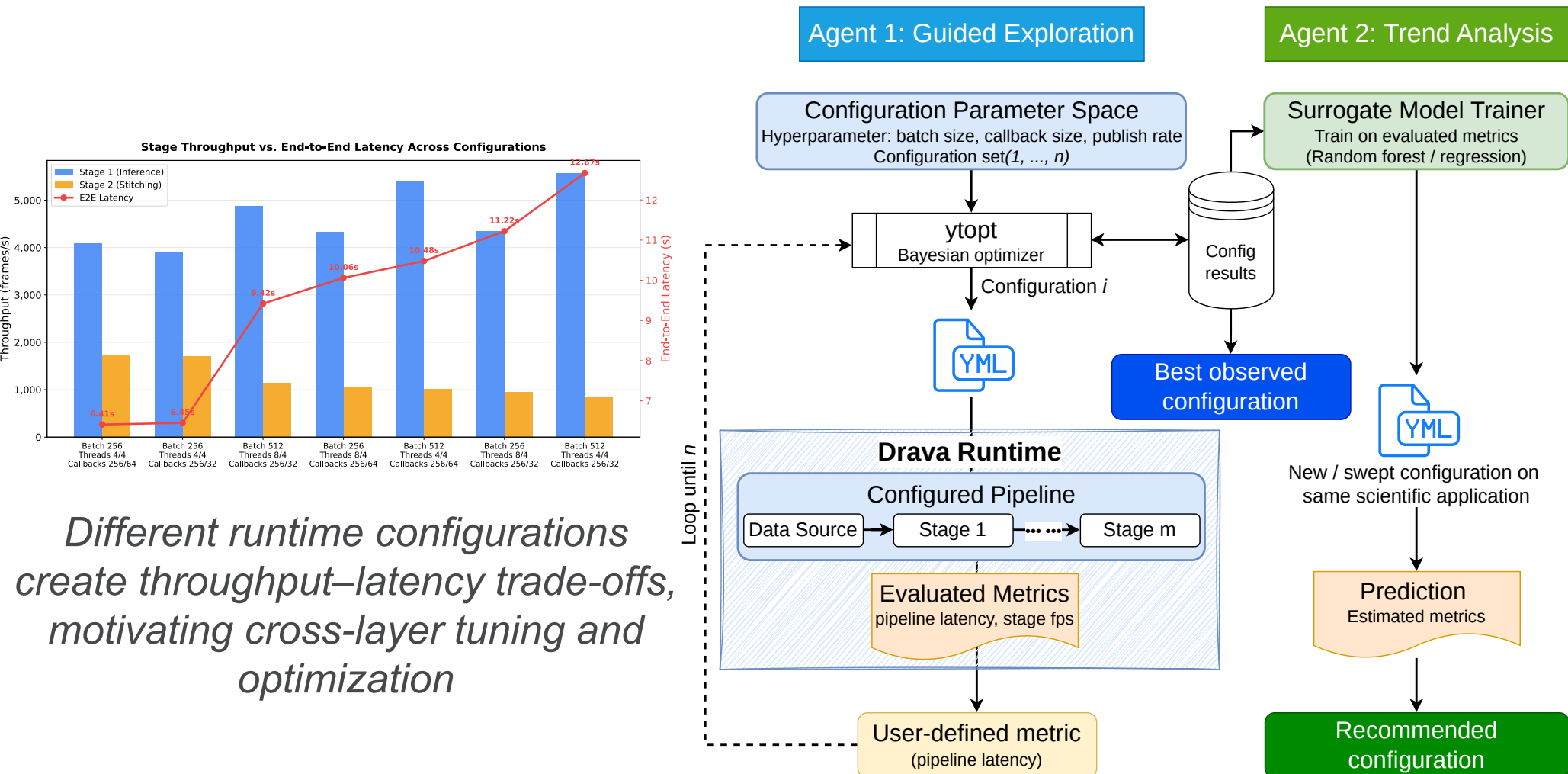
End-to-end application workflow in Drava

Metric	Meaning	Source
rx_msgs	Callback batches processed	Counter
rx_items	Non-EOS frames received	Counter
rx_bytes	Input payload bytes	Counter
tx_msgs	Published payloads	Publish
tx_bytes	Published bytes	Publish
cb_avg_ms	Mean callback-batch duration	Timer
publish_total_s	Cumulative publish time	Timer
stage_avg_ms	Mean dispatch-to-publish time	Samples
stage_total_fps	Frames per stage interval	Snapshot

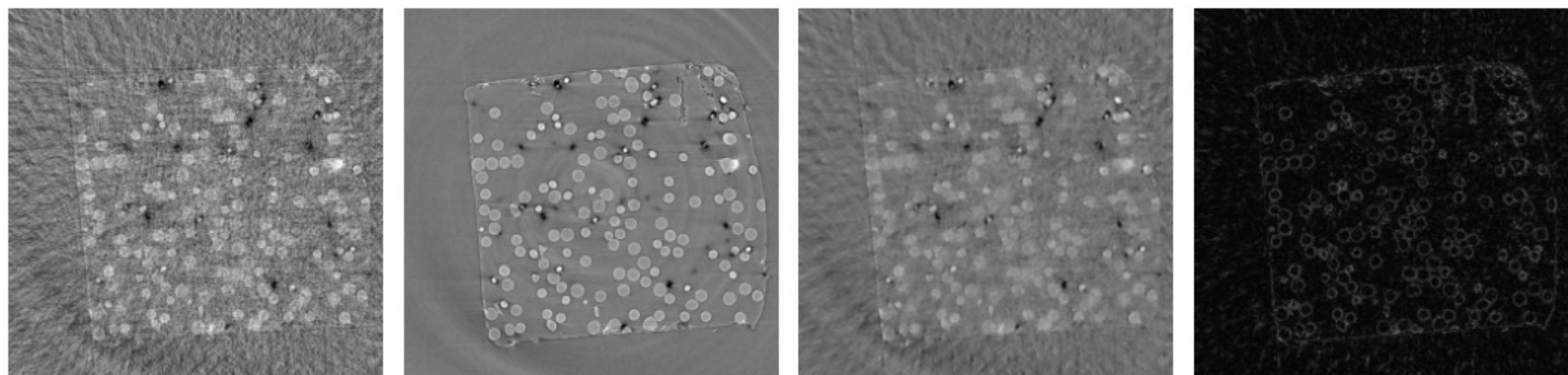
Per-stage metrics reported by Drava

AGENT-AIDED CONFIGURATION OPTIMIZATION

- Drava uses an external agentic workflow to tune runtime and application parameters
- Agent 1 performs guided exploration by launching Drava with selected configurations and recording measured performance
- Agent 2 analyzes the performance database, learns configuration trends, and recommends promising settings. This reduces the need for exhaustive manual tuning in large configuration spaces.



Agentic autotuning of hyperparameter space



TomoGAN inference result: noisy input, ground truth, Drava/TomoGAN denoised output, and absolute error.

CONCLUSIONS

- Drava demonstrates that an event-driven, FaaS-style runtime can sustain multi-stage scientific streaming pipelines while separating application logic from runtime policy through external configuration
- Agentic Bayesian optimization efficiently navigates the combinatorial configuration space, identifying high-performing configurations with fewer evaluations than exhaustive search
- Integrated lock-free instrumentation enables per-stage and pipeline-level performance analysis with negligible overhead on the data path

NEXT STEPS

- Evaluate Drava pipelines across distributed nodes to characterize throughput under network-bound conditions
- Connect runtime metrics to dynamic configuration adaptation, enabling the agentic optimizer to adjust batching and concurrency during execution rather than between runs
- Integrate a high-throughput transport backend and simulate runtime execution on future computational accelerator architectures

REFERENCES

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